ZSWIM8-6 ■■■■ · ■■■■"■■■"■■■■■

■■: Energy-stress–driven metabolic memory via ZSWIM8-dependent miRNA decay ■AMPK ■■■ ZSWIM8-S608 → ■■■■ miRNA ■ TDMD → T2D ■■■■■■■■■ **■■■:** ■FILL: ■■■■ · ■■■ / ■■ · ■■■■■ **■■:** ■■■■■■■■■——■■■■■■■■■■■■■(miRNA ■■ × ■■■■■■■■■)■■,■■■ **■■■■:** 2026-07-22

> **■■■■:** ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■,ZSWIM8-6 ■■■■■■■■■■■ ■`FILL: …`■ ■■■■■■■■■■/■■/■■■■■■■,■■■■■■■■■■■■■■■■■■/■■■■■ predicted/schematic■

■■

- ■■ A · NIH K99/R00■■■→■■■■,■■■■■■■■■■■■■■■
- ■■ B · NIH NRSA F32■■■■/■■,■■■■
- ■■ C · NIH R01■■■ PI ■■,■■■■)
- ■■ D · ■■■■■■■■■■■,■■■■+■■■■)
- ■■ E · ■■ RGC ECS■■■■■■■■+36 ■■■■)
- ■■ F · ■■■ NRF Fellowship■■■■+■■ track record)

■■ A · NIH K99/R00■Pathway to Independence■— ■■■■

■■ NOFO: PAR-25-135■NCI■· ■■■■■■■■ ≤**12** ■■K99 ■■■ 6 ■ + R00 ■■■ 6 ■■· ■■■■ **■■■■:** ■■≤4 ■■■■■■■——■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

SPECIFIC AIMS■1 ■ · ■■■■■

MicroRNAs, once regarded as constitutively stable, are actively destroyed on a physiological timescale by target-directed miRNA decay (TDMD), in which the ZSWIM8 ubiquitin ligase recognizes a highly complementary target engaged by an AGO2–miRNA complex and marks AGO2 for degradation. Although the ZSWIM8 TDMD machinery is now structurally defined, whether and how it is tuned by the cell's metabolic state is unknown. Whether the energy sensor AMPK—activated within minutes of nutrient stress—directly controls the rate of miRNA turnover has never been tested. This gap is pressing because type 2 diabetes (T2D) exhibits "metabolic memory": tissues retain a disordered metabolic program long after glycemic control is restored, yet no molecular mechanism explains this persistence.

Our long-term goal is to define the metabolic control of miRNA stability and exploit it therapeutically. **Our central hypothesis is that energy stress, through AMPK, phosphorylates ZSWIM8 at S608, accelerating TDMD of specific metabolic miRNAs and thereby encoding a durable, reversible metabolic memory. This hypothesis is grounded in our transparent AMPK substrate-motif scan, which places ZSWIM8-S608 in a textbook AMPK recognition context (…KRR-S608…) within a disordered, kinase-accessible region.**

Aim 1 — Test whether AMPK directly phosphorylates ZSWIM8 at S608. In vitro kinase assays with recombinant AMPK and ZSWIM8 fragments, phospho-specific detection, and S608A/S608D mutants will establish direct phosphorylation and its dependence on the predicted motif. Predicted outcome: S608 phosphorylation is AMP-dependent and abolished by S608A.

Aim 2 — Determine how S608 phosphorylation sets the rate of TDMD. Using time-resolved small-RNA sequencing after transcriptional shut-off in ZSWIM8-reconstituted cells (WT / S608A / S608D / KO), we will fit decay-rate constants for candidate miRNAs. Predicted ordering: S608D > WT > S608A > KO.

Aim 3 — Test whether AMPK-driven ZSWIM8 activity encodes metabolic memory. In a T2D pig model and in fibrosis organoids subjected to a defined nutrient-stress pulse, we will ask whether a transient stress produces a durable, S608-dependent shift in metabolic miRNA levels and phenotype. Predicted: transient stress → lasting, S608-dependent miRNA suppression.

Successful completion will establish the first metabolic input to the TDMD machinery, define a phospho-switch (ZSWIM8-S608) as a druggable node, and open metabolic control of miRNA stability as a new axis in diabetes and its complications.

A. CANDIDATE'S BACKGROUND■■■■■■■ · ■ 1 ■■

I trained in biochemistry and molecular biology (PhD, 2022) with a focus on AMPK energy-stress signaling and small-RNA analysis. My seven first-author publications in RNA biology and metabolic disease establish my capacity to lead this interdisciplinary program. ■FILL: ■ 3–4 ■■■■■■■■■ 2 ■■■■■■■■■■■■■■■■■——■■■ RNA ■■■■■■■■■AMPK ■■■■■■■■■■

B. CAREER DEVELOPMENT / TRAINING PLAN■■■■■■■■■■■■ · ■ 1.5 ■■

Objective. To become an independent investigator studying the metabolic control of miRNA stability. **Training goals (mentored K99 phase):** (1) quantitative RNA-decay kinetics and time-resolved small-RNA-seq analysis; (2) in vivo large-animal (T2D pig) metabolic modeling; (3) structural/biochemical reconstitution of the ZSWIM8–AGO2 complex; (4) responsible conduct of research, rigor and reproducibility, and lab-management/mentoring coursework. During the mentored phase I will acquire advanced training in quantitative RNA-decay measurement and in vivo large-animal metabolic modeling—skills I will carry into my independent laboratory. ■FILL: ■■■■■■■■■——■■■■■■■■,■■■■■■■■■■■■■■/■■■■■■■■■ mentored→independent ■■■■■■■■

C. TRAINING ACTIVITIES & MENTORING TEAM■■■■■■■■■■■ · ■ 1 ■■

My mentoring/collaborator team spans TDMD biochemistry, kinase signaling, and organoid disease modeling, covering every methodology in this proposal. ■FILL: ■ 3–4 ■■■/■■■■■■■■■■■■■TDMD ■■■■■■■■■■■■■■■■■ The institutional environment provides ■FILL: ■■■■——■■■■■■■■■■■■■■■■

D. RESEARCH STRATEGY■■■■■ · ■ 6 ■,K99+R00 ■■■

Significance

- Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point.
- Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway.
- Conceptual reach. Because ZSWIM8 governs decay of many miRNAs, a metabolic input would have consequences across the miRNAome, not a single miRNA.
- Unique resources. Direct access to a T2D pig model, intestinal/cardiac organoids, a lab-native small-RNA sequencing pipeline, and base/prime editing positions us to test this axis in human-relevant, large-animal contexts other laboratories cannot easily replicate.

Innovation

- Conceptual: a paradigm shift—miRNA decay is not constitutive but is set by metabolic state through a defined kinase→ligase axis.
- New target: ZSWIM8-S608 as, to our knowledge, the first phosphosite proposed to couple energy sensing to TDMD.
- Methodological: an integrated pipeline—transparent AMPK substrate scorer + endogenous phospho-mimetic editing + single-miRNA decay kinetics—not previously applied to TDMD.
- Editing: base/prime editing installs precise endogenous S608A/S608D alleles, avoiding overexpression artifacts.

Approach — K99 mentored phase■■■■■■■■■■■■■■■■■

Aim 1■Yr 1–2■. Rationale: our motif scan flags S608 (…KRR-S608…) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

Approach — R00 independent phase■■■■■■■■■■■■■■

Aim 2■Yr 3–4■. Design: reconstitute ZSWIM8-KO cells with WT / S608A / S608D; halt transcription; perform a time-course small-RNA-seq. Readout: per-miRNA decay-rate constant k. Predicted result (schematic): S608D > WT > S608A > KO. Statistics: powered to detect a 30% difference in k at 80% power (α=0.05), ≥3 biological replicates. Pitfall & mitigation: to separate global from selective effects, normalize to TDMD-insensitive miRNAs.

Aim 3■Yr 4–5■. Design: (i) T2D pig tissue under a controlled metabolic challenge; (ii) intestinal/cardiac fibrosis organoids given a defined nutrient-stress pulse and tracked for 7–14 days, with endogenous S608A/S608D installed by prime editing. Readout: durable change in metabolic miRNA levels and phenotype after stress withdrawal. Predicted result (schematic): transient stress → lasting, S608-dependent suppression (metabolic memory). Alternative: if organoids lack durability, extend the pulse or apply repeated pulses.

Rigor & reproducibility: both sexes will be studied and sex included as a biological variable; all antibodies, cell lines and ZSWIM8 constructs authenticated and reported per RRID standards; a go/no-go milestone requires Aim 1 to confirm S608 phosphorylation before Aim 2 kinetics proceed. The independent phase establishes a program on the metabolic control of miRNA stability that is distinct from my mentor's focus on TDMD structure, giving me clear intellectual ownership.

■■■■■■:K99 ■■■ Aim 1(■■■■■■■■■■■);R00 ■■ Aim 2–3(■■■■■■■■■)■■■■■■■■■■■■■——■■ K99/R00 ■■■■■■■

■■■■■K99/R00 ■■■

- [ ] Aim 1 ■ K99■Aim 2–3 ■ R00,■■■■■

- [ ] ■■■■■■■■■■/■■/■■■■■■
- [ ] ■■■■■■■■■■■■■
- [ ] SABV(■■)+ RRID ■■■■■■■ Approach
- [ ] ■■■■■■■■■■■■■ predicted/schematic
- [ ] ■■:■■■■ ≤12 ■;■■/■■■■ SF424 Guide

■■ B · NIH NRSA F32■■■■■ Fellowship■— ■■■■

■■■■■■ ≤6 ■ · ■"■■"■"■■" · ■■■■ ■■ ■■: F32 ■■■■■■/■■——■■■■■■■ NOFO ■ citizenship ■■,■■■■■ K99■

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Successful completion will establish the first metabolic input to the TDMD machinery, define a phospho-switch (ZSWIM8-S608) as a druggable node, and open metabolic control of miRNA stability as a new axis in diabetes and its complications.

■F32 ■■:■■■■■ Aim 1–2(■■■■■■■■■■■),Aim 3 ■■"future direction"■■■■,■■■■■■■■■

RESEARCH TRAINING PLAN■■■■■■■ · ■ 3–4 ■■

Background & Significance■■■■■■■■

Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point. Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway.

Approach

Aim 1. Rationale: our motif scan flags S608 (…KRR-S608…) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

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TRAINING GOALS & OBJECTIVES■■■■■——F32 ■■■,■ 1 ■■

What I will learn that I cannot do now: (1) quantitative miRNA-decay kinetics; (2) time-resolved small-RNA-seq experimental design and analysis; (3) rigorous biochemical reconstitution. **How this training advances my path to independence:** During the mentored phase I will acquire advanced training in quantitative RNA-decay measurement and in vivo large-animal metabolic modeling—skills I will carry into my independent laboratory. ■FILL: ■■■■■■■■■■■■■■■■■■/■■,■■■■■■■■■■■■F32 ■■■■■■■■"■■■■",■■"■■■■■"■■

SPONSOR(S) & TRAINING ENVIRONMENT■■■■■■■■■ · ■ 1 ■■

My mentoring/collaborator team spans TDMD biochemistry, kinase signaling, and organoid disease modeling, covering every methodology in this proposal. ■FILL: ■■■■■■■■■■■■■■■■■■■■■■;■■■■■■■■■

RESPONSIBLE CONDUCT OF RESEARCH■RCR · ■■,■■■

■FILL: ■■■ RCR ■■/■■■■■■■■■——F32 ■■■■■■■■■■■

■■■■■F32 ■■■

- [ ] ■■■ citizenship ■■
- [ ] ■■ Aim 1–2;Aim 3 ■■■■
- [ ] "■■■■"■■■■,■■■■■■■+■■
- [ ] RCR ■■■■
- [ ] SABV + RRID ■■■
- [ ] ■■■■ ≤6 ■

■■ C · NIH R01■■■■■■■■■— ■■■■

■■■■■■ ≤**12** ■ · ■■ Career ■ · ■■■■ **■■■■:** ■■■■■■■■■■ PI——■■■■"■■■■"■■,K99→R00→R01 ■■■■■■■;■■■■■■■■■■■■■■■

SPECIFIC AIMS■1 ■■

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RESEARCH STRATEGY■■■■■ · ■ 11 ■■

Significance■■ 2 ■■

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- Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway.
- Conceptual reach. Because ZSWIM8 governs decay of many miRNAs, a metabolic input would have consequences across the miRNAome, not a single miRNA.
- Unique resources. Direct access to a T2D pig model, intestinal/cardiac organoids, a lab-native small-RNA sequencing pipeline, and base/prime editing positions us to test this axis in human-relevant, large-animal contexts other laboratories cannot easily replicate. ■FILL: R01 ■ Significance ■■ fellowship ■■——■■■■■■■■■■ + ■■■■■■■■,■ 8–12 ■■■■■

Innovation■■ 1 ■■

- Conceptual: a paradigm shift—miRNA decay is not constitutive but is set by metabolic state through a defined kinase→ligase axis.
- New target: ZSWIM8-S608 as, to our knowledge, the first phosphosite proposed to couple energy sensing to TDMD.
- Methodological: an integrated pipeline—transparent AMPK substrate scorer + endogenous phospho-mimetic editing + single-miRNA decay kinetics—not previously applied to TDMD.
- Editing: base/prime editing installs precise endogenous S608A/S608D alleles, avoiding overexpression artifacts.

Approach■■ 7–8 ■,■■■■■■■

Preliminary Data. ■FILL: R01 ■■■■■■■■■■■■ AMPK ■■ PSSM ■■■■■S608 ■■■■■,■■■■■■■ ZSWIM8/TDMD ■■■■ predicted/schematic ■■■■■;R01 ■■■"■■■"■■■■■■ fellowship,■■■■■■■■■■

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■■■■■R01 ■■■

- [ ] ■■■ PI ■■(■■■■■■■■■)
- [ ] Significance ■■■■■ + ■■■■
- [ ] Preliminary Data ■■,■■■■
- [ ] ■ Aim ■ Expected/Pitfalls/Alternatives ■■
- [ ] SABV + RRID + Authentication ■■
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■■■■■■■"■■"■■■ miRNA ■■■■■■■■■■■■■?■■■■:AMPK ■■■■■ TDMD ■■■■■■ ZSWIM8(S608),■■■■■■■■ miRNA ■■■■■■■(TDMD),■■■■■■■■■■■■■■■■■■■■■■■■ ■■"■■–TDMD"■■■■■■■■

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Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point. Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway. Conceptual reach. Because ZSWIM8 governs decay of many miRNAs, a metabolic input would have consequences across the miRNAome, not a single miRNA.

■■■■■■■: ■FILL: ■■■■■■■■■■■——■■■■■■■■■■■■;■■■■■■/■■/■■■■;■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

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■■■■■■■■■■■■,≤5 ■■

■FILL: ■ 5 ■■■,■■■■■■/■■/■■■■/■■■■/■■;■■■■■■■■(■■×miRNA)■■■■■

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■■■■■(T2D ■)■■■■■■■■■■;■■■■■■■■■■■■■■,■"■■■■■"■■■■■■■■■■FILL: ■■■■■■■■■■■■■■■5 ■■■■■■■■■■■■■■■■

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■■ E · ■■ RGC Early Career Scheme (ECS) — ■■■■■■■■■■■

■■: Objectives → Background of Research → Research Plan & Methodology → Impact & Significance **■■:** ■■■■■■■■■■■ AP;■■■■■■ R01,■"■■■ + ■■■■"■

OBJECTIVES■■■■■■■■■■■■■■

1. Determine whether AMPK directly phosphorylates ZSWIM8 at S608 and define the kinase–substrate relationship.
2. Establish whether S608 phosphorylation accelerates TDMD of a defined set of metabolic miRNAs (keystone: miR-29).
3. Test whether this axis encodes a reversible "metabolic memory" in a disease-relevant model.

BACKGROUND OF RESEARCH

Clinical burden. Metabolic memory drives the persistence of diabetic complications despite glycemic control; no therapy erases this memory. A metabolic lever on miRNA decay would be a mechanistically novel entry point. Mechanistic advance. Establishing AMPK as an upstream regulator of TDMD would convert miRNA turnover from a housekeeping process into a signal-responsive, druggable pathway. Unique resources. Direct access to a T2D pig model, intestinal/cardiac organoids, a lab-native small-RNA sequencing pipeline, and base/prime editing positions us to test this axis in human-relevant, large-animal contexts other laboratories cannot easily replicate. ■FILL: ECS ■■■■,■■ ≤1.5 ■,■■"gap → ■■■■"■■

RESEARCH PLAN & METHODOLOGY

Objective 1. Rationale: our motif scan flags S608 (…KRR-S608…) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

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Objective 3. Design: (i) T2D pig tissue under a controlled metabolic challenge; (ii) intestinal/cardiac fibrosis organoids given a defined nutrient-stress pulse and tracked for 7–14 days, with endogenous S608A/S608D installed by prime editing. Readout: durable change in metabolic miRNA levels and phenotype after stress withdrawal. Predicted result (schematic): transient stress → lasting, S608-dependent suppression (metabolic memory). Alternative: if organoids lack durability, extend the pulse or apply repeated pulses.

Rigor & reproducibility: both sexes will be studied and sex included as a biological variable; all antibodies, cell lines and ZSWIM8 constructs authenticated and reported per RRID standards; a go/no-go milestone requires Aim 1 to confirm S608 phosphorylation before Aim 2 kinetics proceed. ■FILL: ECS ■■■■"36 ■■■■■■"——■■■■■■■■■■■■■■,■■■■■■■■■■■■■/■■■■■

IMPACT & SIGNIFICANCE

Conceptual: a paradigm shift—miRNA decay is not constitutive but is set by metabolic state through a defined kinase→ligase axis. Editing: base/prime editing installs precise endogenous S608A/S608D alleles, avoiding overexpression artifacts. ■FILL: ■■■■■/■■■■■■■■■■■■,■■■■■■■■■■■■■

■■■■■RGC ECS ■■■

- [ ] Objectives ■■■■■■■■■
- [ ] ■■ ≤1.5 ■,■■ gap
- [ ] 36 ■■■■■ + ■■■■
- [ ] Impact ■■■■■
- [ ] ■■■■■■■ AP offer

■■ F · ■■■ NRF Fellowship — ■■■■■■■■■■

■■: Research Vision → Track Record → Research Plan → Budget & Host Institution **■■:** ■■■■■ fellowship,5 ■■■■■;■■"■■■■■■■ + ■■ track record"■

RESEARCH VISION■■■■■——NRF ■■■,■■■■■

I will establish that **cellular energy state is transduced into durable, reversible programs of miRNA turnover** — a new layer of metabolic memory — and build an independent program mapping the "kinase–TDMD" interface across metabolic disease. Successful completion will establish the first metabolic input to the TDMD machinery, define a phospho-switch (ZSWIM8-S608) as a druggable node, and open metabolic control of miRNA stability as a new axis in diabetes and its complications. ■FILL: NRF ■ 10 ■■■■■■,■■■■■■;■■■■■■■"■■■"■■

TRACK RECORD■■■■■■■■■■■■

■FILL: ■■■■■ + ■■■■■■■■■ + ■■■■■■■■ + ■■■■■■NRF ■■■ PI ■■■■,■■■■■(■■■■■■■■■■■■■■)■■

RESEARCH PLAN

Aim 1. Rationale: our motif scan flags S608 (…KRR-S608…) in a disordered, accessible region. Design: recombinant AMPK + ZSWIM8-fragment kinase assays ± AMP; phospho-specific antibody/mass spectrometry; S608A and S608D point mutants. Readout: stoichiometry of S608 phosphorylation. Predicted result (schematic): AMP-dependent phosphorylation, abolished by S608A. Alternative: if S608 is insufficient, test the neighboring S609 and S1202 sites (also flagged by our scan) and screen AMPK-independent kinases. Pitfall & mitigation: S608D may not fully mimic phospho-S608; corroborate with in-vitro-phosphorylated protein and phospho-specific antibodies.

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Rigor & reproducibility: both sexes will be studied and sex included as a biological variable; all antibodies, cell lines and ZSWIM8 constructs authenticated and reported per RRID standards; a go/no-go milestone requires Aim 1 to confirm S608 phosphorylation before Aim 2 kinetics proceed.

BUDGET & HOST INSTITUTION

■FILL: 5 ■■■■■(■■/■■/■■/■■)+ ■■■■■■■■■(■ A*STAR/NUS/NTU)■■■■■■■ + ■■■■■■

■■■■■NRF ■■■

- [ ] ■■■ 10 ■■■■■■
- [ ] Track record ■■■■■■■■
- [ ] 5 ■■■ + ■■■■■■■
- [ ] ■■■■■■ host ■■